MIL-B-52203D
12 November 1974
SUPERSEDING
MIL-B-52203C
1 April 1969

MILITARY SPECIFICATION

BOOM CRANE: 20-TON, ROUGH-TERRAIN; 20-TON, TRUCK-MOUNTED; 12-1/2-TON, CRAWLER-MOUNTED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a military standard boom with extensions and attachments for a 20-ton rough terrain crane, a 20-ton truck crane and a 12-1/2-ton crawler crane.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-3580	- Cranes and Crane-Shovels, Truck, Crawler, and Wheel Mounted, Full Revolving, and Their Attachments, Packaging of.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-B-52207	- Boom Jib, Crane, 15-Foot-Long, for 3/4-Cubic-Yard (12-1/2-Ton Crawler Mounted, 20-Ton Wheel Mounted) Cranes.

STANDARDS

Military

MIL-STD-130	- Identification Marking of US Military Property.

FSC 3815

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DRAWINGS

ME

TA13217E6260 - Boom, Crane: 20-Ton Truck-Mounted; 12-1/2-Ton Crawler-Mounted.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Welding Qualifications, Section IX.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY, INC. (AWS)

D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 NW Seventh Street, Miami, FL 33125.)

3. REQUIREMENTS

3.1 Description. The basic boom shall consist of one inner boom section and one outer boom section with pendant cables. The boom shall be in accordance with Top Assembly TA13217E6260 covering a boom for a crane capable of lifting 40,000 pounds. The basic boom shall be 30 feet in length; a maximum length of 90 feet shall be attainable by inserting six 10-foot extensions. A 15-foot jib boom conforming to MIL-B-52207 may be used when the boom is assembled to 80 feet or less. A fairlead may be used with the assembled boom when required.

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3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the supplier shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the booms. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The supplier shall furnish a boom as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that his production methods will produce booms that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified in TA13217E6260 and as specified herein.

3.4 Boom extensions. When specified (see 6.2) boom extensions conforming to Drawing 13217E6263 shall be furnished. One pair of matched pendant cables conforming to Drawing 13217E6266 shall be furnished with each boom extension.

3.5 Fairlead. When specified (see 6.2) fairleads conforming to Drawing 13217E6261 shall be furnished.

3.6 Delivery lubrication service. When delivered, all components requiring lubrication shall be lubricated. Grease lubrication shall be with greases conforming to MIL-G-10924. Each boom shall be tagged in a conspicuous place to indicate that it has been lubricated.

3.7 Identification marking. The inner and outer boom sections, fairlead, and boom extensions each shall be identified in accordance with MIL-STD-130 on plates conforming to Drawing 13217E6281.

3.8 Treatment and painting. The portions of the boom and its components and parts normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, Type A, color as specified (see 6.2).

3.9 Government-loaned property. When specified (see 6.2), the following property in the quantities indicated will be loaned by the Government (see 6.4):

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Item No.	Description	Identification	Qty for each boom
1	Boom, jib, crane, 15-foot-long, for 3/4-cubic-yard crane-shovel.	NSN 3815-00-542-4839	1
2	Fairlead, roller and sheave.	NSN 3815-00-554-0126	1
3	Boom extension, middle, crane.	NSN 3815-00-540-4756	as required.

3.10 Workmanship. The boom shall be free from defects such as incomplete welds, rust, cracks, and other defects that could impair the operation or serviceability of the boom.

3.10.1 Steel fabrication. Steel used in the fabrication of the boom shall provide original surface finish and shall be free from kinks and sharp bends. Straightening of material shall be done by methods that will not cause injury to the metal. Straightening of boom chord angles shall not be permitted either before or after fabrication. Shearing and chipping shall be done neatly and accurately. Corners shall be square and true, and all sharp edges and burrs shall be removed. Flame cutting may be employed instead of shearing or sawing. Burned surfaces of flame-cut material shall be ground or machined sufficiently to remove ash and cooling checks. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly. All bends of a major character shall be made with metal dies or fixtures in order to insure uniformity of size and shape.

3.10.2 Bolted connections. Boltholes shall be accurately formed or drilled and burrs and sharp edges shall be removed. All bolts, nuts, and screws shall be correctly torqued. Adjoining surfaces shall be flat and in full contact with each other.

3.10.3 Welders and welding.

3.10.3.1 Welders. Before assigning any welder to manual welding work covered by this specification, the supplier shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

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AWS D1.1, Structural Welding Code, Section 5, Qualification.
Welding Qualifications of the ASME, Section IX.

Suppliers who only make horizontal welds need not qualify welders for "all position welding." Subject to approval by the Government, supplier's standard welder qualification may be substituted in lieu of the above codes provided that the supplier's procedure is equivalent to the above codes. The supplier shall be responsible for determining that automatic welder equipment operators are capable of producing quality welds in accordance with AWS and ASME codes.

3.10.3.2 Welding. The surfaces of parts to be welded shall be free from scale, paint, grease, and other foreign matter. When multiple layers of weld metal are required, each layer shall be thoroughly cleaned before depositing another layer. All welded parts and assemblies shall be free from cracks and other imperfections that may reduce the effectiveness of the part or assembly. All weldments shall be free from slag, flux, weld spatter, and other impurities. Work shall be positioned for flat welding whenever practicable. No undercuts or overlays will be permitted on any weldment. Butt type joints having members of equal thickness shall be alined within 10 percent of the thickness of the members involved. Where different steels are welded to each other, filler metal having strength of the lower strength steel shall be used. When the same steels, or steels having similar properties are used, the deposited filler metal shall develop the full design strength of the particular weld joint or weldment.

3.10.3.3 Mechanical properties. Mechanical properties of all welds shall be such that welds will transmit stress without failure or permanent deformation when subjected to any test specified in this specification.

3.10.4 Machine work. Tolerances and gages for metal fits shall conform to the limitations specified on drawings. All holes shall be countersunk before tapping.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.1.2 Disassembly inspection. Failure of any examination or test by the preproduction model shall be cause for disassembly, in the presence of a Government representative, of the preproduction model to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the preproduction model. Reassembly with replacement parts and retesting shall be the responsibility of the supplier.

4.1.3 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with Supplementary Quality Assurance Provisions (SQAP) shown on the drawings. The drawings specify the characteristics requiring SQAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.5).

4.2 Classification of inspections. Inspections shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction inner and outer boom sections, fairlead, and boom extension shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Test. The preproduction inner and outer boom sections, fairlead, and boom extension shall be tested as specified in 4.5.2. Failure of any test shall be cause for performing the inspection specified in 4.1.2.

4.4 Quality conformance inspection.

4.4.1 Examination. Each inner and outer boom section, fairlead, pendant cable, and boom extension shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

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4.4.2 Tests. Each fairlead and pendant cable shall be tested as specified in 4.5.2.5 and 4.5.2.6. Failure of the test shall be cause for rejection.

4.5 Inspection procedure. The inner and outer boom sections, fairlead, and boom extensions shall be inspected as applicable.

4.5.1 Examination. The boom shall be examined for the following defects:

101. Size of holes and hole pattern in each pad of basic boom and extension section not as specified.
102. Distances between diagonally opposite pads of each connection point of the basic boom and extension sections not within drawing tolerance.
103. Distance between adjacent pads not as specified.
104. All pads at each end of the basic boom and extension sections not in same plane within drawing tolerance.
105. Connection bushing bore of inner boom section not as specified.
106. Sheave pin bushing bore of fairlead not as specified.
107. Sheave pin bushing bore of outer boom section not as specified.
108. Size or location of holes in jib boom anchor plates on outer boom section not as specified.
109. Other dimensions not as specified.
110. Assembly not complete or not correct.
111. Parts or components missing.
112. Material not as specified.
113. Drill hole, spotface, tap hole, and threaded bushing for lubricating fitting in boom foot hub not as specified on Drawing 13217E6262.
114. Capscrews and lockwashers not as specified for fairlead mounting pad.
115. Pendant cables not furnished in matched pairs.
116. Fairlead not constructed or alined as specified.
117. Wire rope not as specified.
118. Lubrication fitting missing.
119. Identification or other marking missing.
120. Treatment and painting not as specified.
121. Welding not as specified.
122. Workmanship not as specified.

4.5.2 Tests.

4.5.2.1 Assembly. Assemble a 60-foot boom using a basic boom, three boom extensions, a fairlead, and a 15-foot jib boom without the use of

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jacks or clamps, and without modification of any component. Inability to assemble the boom with extensions or attachments, or evidence of unusual force required to aline components shall constitute failure of this test. Boom extensions, a fairlead, and a jib boom will be loaned by the Government (see 3.9) if not otherwise a part of the procurement.

4.5.2.2 Sheave and roller test. After the boom has been assembled as specified in 4.5.2.1, rotate all rollers and sheaves. Misalinement or binding shall constitute failure of this test.

4.5.2.3 Basic boom. When basic booms only are furnished by the supplier (outer and inner sections), test the boom as shown in Figure 1. Secure the boom foot to an A-frame with simulated boom-foot brackets. The centerline of the sheave on the A-frame shall be 54-11/16 inches plus or minus 1/16 inch vertically above, and 79 inches plus or minus 1/16 inch horizontally back of the boom-foot brackets. Raise the boom until the load-contact point of the hook is on a radial line not less than 20 feet, 4 inches horizontally from the center of the boom-foot pin. Attach a load of 40,000 pounds to the hook and raise the load not less than 1 foot from the ground. Sustain the load in this position for not less than 10 minutes before lowering the load to the ground. Repeat this test 10 times. Permanent deformation, weld failure, or inability to raise and hold a load of 40,000 pounds for 10 minutes shall constitute failure of this test.

4.5.2.4 Basic boom and extensions. When both basic boom and extensions are furnished by the supplier, assemble the boom and extensions to a total length of 60 feet. The boom shall be rigged, and the boom foot shall be secured to an A-frame as specified in 4.5.2.3 and as shown in Figure 1. Raise the boom until the load contact point of the hook is on a radial line not less than 14 feet, 4 inches horizontally from the center of the boom-foot pin. Attach a load of 28,000 pounds to the hook and raise the load not less than 1 foot from the ground. Sustain the load in this position not less than 10 minutes before lowering the load to the ground. Repeat this test 10 times. Permanent deformation, weld failure, or inability to raise and hold a load of 28,000 pounds for 10 minutes shall constitute failure of this test.

4.5.2.5 Pendant cable proof test. Measure each pendant cable while under a load of 45,000 pounds for variations in length of cables shown on Drawings 13217E6265 and 13217E6266. Select matched pairs of cables and secure them

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together. Damage to or permanent deformation of the pendant cables, length not as specified on the drawings, or a maximum variation of length between pendant cables in a matched pair of more than specified shall constitute failure of this test.

4.5.2.6 Fairlead. Rotate all rollers and sheaves of the fairlead. Misalinement or binding shall constitute failure of this test.

4.6 Inspection of preparation for delivery. The preservation, packaging, packing, and marking shall be examined to determine compliance with the Quality Assurance Provisions of MIL-C-3580.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Each complete boom shall be preserved, packaged, packed, and marked in accordance with MIL-C-3580. The level of preservation and packaging shall be Level A, B, or C and the level of packing shall be Level A, B, or C as specified (see 6.2).

6. NOTES

6.1 Intended use. The boom is intended for use on military 20-ton, rough terrain cranes, 20-ton truck cranes, and 12-1/2-ton crawler cranes.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of preproduction model (see 3.2).
(c) When boom extensions are required (see 3.4).
(d) When fairleads are required (see 3.5).
(e) Color required (see 3.8).
(f) When Government-loaned property will be furnished (see 3.9).
(g) Level of preservation and packaging and level of packing required (see 5.1).

6.3 Preproduction model. Any changes or deviations of production booms from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish booms conforming to this specification.

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6.4 Government-loaned property. The contracting officer should arrange to loan the property specified in 3.9 when applicable (see 6.2).

6.5 Supplementary quality assurance provisions (SQAP). The contracting officer should require the supplier to maintain records of all SQAP inspections. A suggested paragraph is as follows:

> "The supplier shall maintain complete records of all examinations and tests performed to verify the requirements of classified SQAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, corrective action taken when applicable."

6.6 This document was prepared by the USAMERDC for USATACOM.

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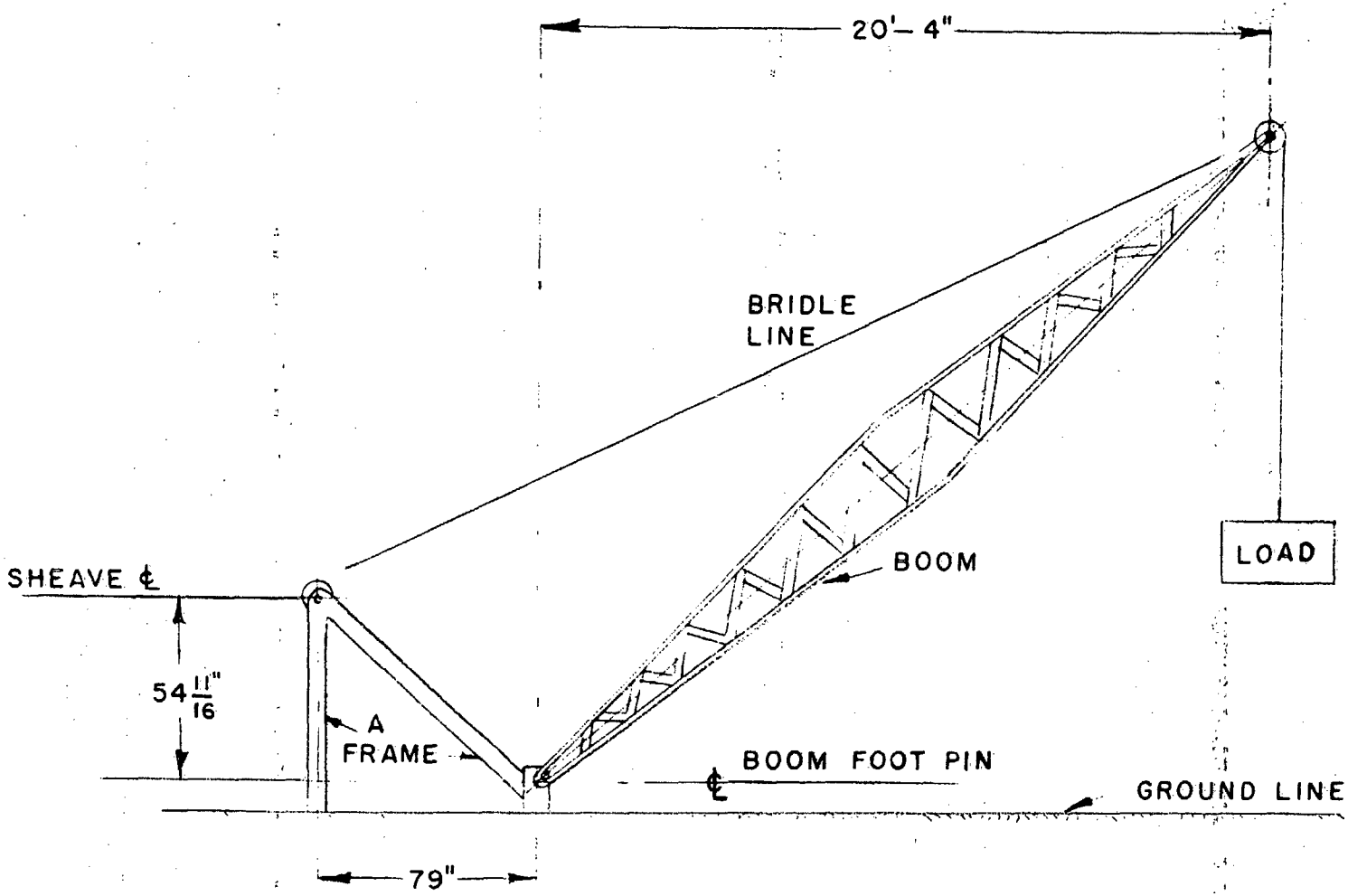


FIGURE 1. BOOM TEST DIAGRAM X1649

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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DOCUMENT IDENTIFIER AND TITLE: MIL-B-52203D - BOOM CRANE: 20-TON, ROUGH-TERRAIN; 20-TON, TRUCK-MOUNTED: 12-1/2-TON, CRAWLER-MOUNTED

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes"; in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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